# Bayesian Statistics III: Bayesian Linear Regression

## ECON 8310: Business Forecasting — Lecture 12

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# Lecture Outline

1 Bayesian Linear Regression

2 Reading the Posterior

3 Causal Structure and DAGs

4 Course Synthesis and Next Steps

## Bayesian Linear Regression

The same regression you already know — returning a distribution for every coefficient.

## OLS vs. Bayesian Regression: What Changes?

OLS, which you have fitted since Lecture 2, solves for one best coefficient vector:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

The Bayesian version uses Lecture 10's update rule instead:

$$p(\boldsymbol{\beta} \mid \mathbf{y}) \;\propto\; p(\mathbf{y} \mid \boldsymbol{\beta}) \cdot p(\boldsymbol{\beta})$$

**Every coefficient comes back as a distribution**: not an estimate plus an asymptotic error bar, but a full posterior you can ask questions of.

And Lecture 6 reappears: **regularization is a prior**. The posterior *mode* under a Normal prior is the Ridge estimate; under a Laplace prior, it is the LASSO estimate.

# Bayesian Linear Regression in PyMC

## Log units on log price, SNAP, events, series and trend

```
with pm.Model() as model:
  a       = pm.Normal("a", y.mean(), 1, shape=J)  # one per series
  b_price = pm.Normal("b_price", 0, 1)     # weakly informative
  b_snap  = pm.Normal("b_snap", 0, 1)
  b_event = pm.Normal("b_event", 0, 1)
  b_trend = pm.Normal("b_trend", 0, 1)
  sigma   = pm.Exponential("sigma", 1.0)
  mu = a[idx] + b_price*lpc + b_snap*snap + b_event*event + b_trend*yr
  pm.Normal("obs", mu=mu, sigma=sigma, observed=y)
  idata = pm.sample(1000, tune=1500, chains=4)
```

**Do not standardize log price** — with both in logs `b_price` is an **elasticity**, percent change in demand per percent change in price. `lpc` is log price centered within series, which fixed ESS 184 $\rightarrow$ 3,097. `J` is the number of series (30), `idx` maps each row to its series, and `yr` counts years since the first week.

## Reading the Posterior

Three questions you can ask a distribution that you cannot ask a point estimate.

## Reading a Posterior Coefficient

Fitted on the weekly panel, controlling for series and trend:

| Coefficient | Mean | 94% HDI | Reading |
|---|---|---|---|
| `b_price` | $-0.493$ | $[-0.596, -0.387]$ | Inelastic: all of it above $-1$ |
| `b_trend` | $0.110$ | $[0.107, 0.113]$ | About 11% growth per year |
| `b_snap` | $0.036$ | $[0.027, 0.046]$ | About a 3.7% SNAP-week lift |
| `b_event` | $-0.003$ | $[-0.011, 0.004]$ | **Spans zero** |

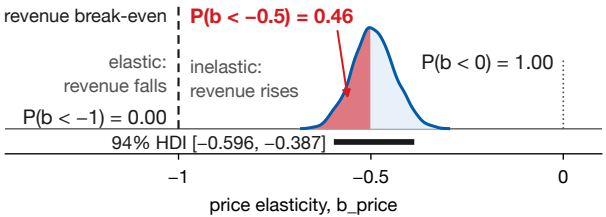
The **HDI** — highest density interval — is the shortest interval holding 94% of the posterior, and it means what everyone wants: a 94% probability the coefficient is inside.

**The last row is the useful one.** `b_event`'s interval spans zero, so the data do not establish a direction. Two slides on, we ask whether it is also too small to matter.

# Asking the Posterior a Direct Question

## Any question about $\beta$ is a count of samples: one line each

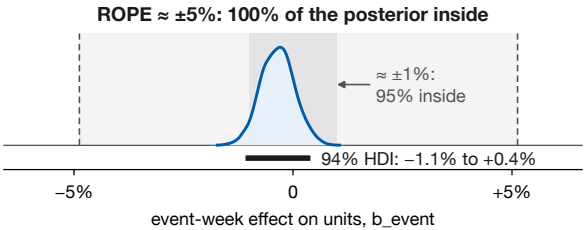
```
b = idata.posterior["b_price"].values.ravel()
(b < 0).mean()      # P(demand falls when price rises)
(b < -1).mean()     # P(elastic: a price rise loses revenue)
(b < -0.5).mean()   # P(a 1% rise costs over 0.5% of volume)
```




Each answer is the share of the area past a line. Demand is **inelastic** in every draw; the half-percent question stays open. **The usual p-value, against zero, gives only the direction.**

# Too Small to Matter? A ROPE

`b_event`'s HDI spans zero: no direction. A **ROPE** — region of practical equivalence — asks a different question: is the effect too small to act on?




The width is a business choice, not a statistic. We use Homework 7's $|\beta| < 0.05$, about $\pm 5\%$ a week; even at $\pm 1\%$, 95% is inside.

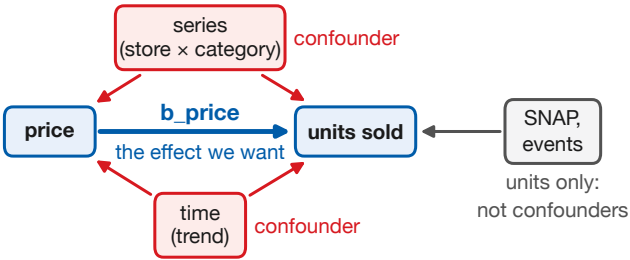
Spans zero and inside the ROPE: **negligible**, stop looking. Spans zero and wide: **still open**, get more data. "Not significant" cannot tell them apart.

## Causal Structure and DAGs

Which coefficients mean what you think they mean.

# Directed Acyclic Graphs

A **DAG** is a picture of what you believe causes what, drawn from domain knowledge *before* the model. It decides which variables belong in the regression.




**Confounder**: causes both predictor and outcome. **Must** be controlled, or it leaks into your coefficient.

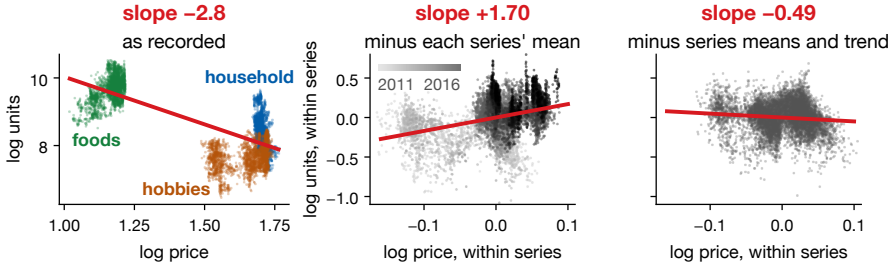
**Collider**: caused by both. Controlling for it **creates** a spurious association (McElreath 2020).

**So "add every control you have" is bad advice**, and no model fit settles it.

*Whether a DAG is right is a business question, not a statistical one.*

# Confounding, Measured

Two confounders move price and units together: the **category** and **time**.




Food is cheap and sells in volume; prices and volumes both rose, 2011–2016. Removing the series means alone flips the sign. With both removed: $-0.493$, HDI $[-0.596, -0.387]$. The naive $-2.805$, HDI $[-2.856, -2.756]$, is nearly six times too large, at half the width.

**Confidently, precisely wrong**: a tight interval says the model is sure, not that it answers your question. Only the DAG protects you from an omitted confounder.

## Scenario Analysis with Posterior Samples

Push a scenario through the draws to get a **distribution of outcomes**, not one number.

**What if we raise price 5%?**

```
b = idata.posterior["b_price"].values.ravel()
pct_change = 100 * (np.exp(b * np.log(1.05)) - 1)
pct_change.mean(), np.percentile(pct_change, [3, 97])
```

With an elasticity near $-0.493$, a 5% price rise implies roughly a **2.4% fall in units**, 94% of the posterior between $-2.9\%$ and $-1.9\%$.

**The decision follows.** Revenue changes by a factor $1.05^{1+b}$, which rises exactly when $b > -1$. Every draw is above $-1$, so $P(\text{revenue rises}) = 1.00$: about $+2.5\%$ on average.

*Valid only to the extent the DAG is — this is a causal claim about intervening on price.*

## Course Synthesis and Next Steps

From classical time series to Bayesian hierarchical models: a complete forecasting toolkit.

# Course Method Map

| Method | Data | Uncertainty | Interpretable | Strengths |
|---|---|---|---|---|
| ETS / ARIMA (L01–02) | Univariate TS | CI | High | Fast, automatic |
| GAM / Prophet (L03) | TS + predictors | Limited | High | Nonlinear + interpretable |
| Decision Tree (L04) | Tabular | No | High | Rule-based explanation |
| Random Forest (L05) | Tabular | OOB error only | Medium | Robust, minimal tuning |
| XGBoost (L05) | Tabular | No | Low | Interactions; 781 on our board |
| Ridge / LASSO (L06) | Tabular | No | High | Best on our board: 744 |
| FFN / 1D CNN (L07–08) | Tabular / TS | No | Low | Flexible nonlinear fits |
| LSTM / Transformer (L09) | Sequences | No | Very low | Long-range dependencies |
| Bayesian (L10–12) | Any | Full posterior | High | Uncertainty, small samples |

No single method dominates. Choose based on: (1) data type and size, (2) need for uncertainty quantification, (3) need for interpretability, (4) time to deploy. In practice, combine: your scoreboard's winner for the point forecast (here, LASSO) and a Bayesian model for the intervals.

# Lecture 12: Key Takeaways

1. **Bayesian regression**: a posterior per $\beta$, whose mode is Ridge (Normal) or LASSO (Laplace).
2. **Ask the posterior directly**: mean, HDI, $P(\beta_j$ past a threshold), mass in a ROPE.
3. **Draw the DAG first**: control confounders (here, series *and* time), never colliders.
4. **Scenarios**: push a counterfactual through the draws to get a distribution of outcomes.
5. **No single method is best**: pair an accurate point forecast with Bayesian intervals.

# References I


McElreath, Richard (2020). *Statistical Rethinking: A Bayesian Course with Examples in R and Stan*. 2nd. Boca Raton, FL: Chapman and Hall/CRC.